

NumPy for Machine Learning — Quick Reference

Function / Method	Syntax	Simple explanation
import numpy	import numpy as np	Imports NumPy with the name np.
array()	np.array([1, 2, 3])	Creates a NumPy array.
zeros()	np.zeros(5)	Creates an array filled with zeros.
ones()	np.ones(5)	Creates an array filled with ones.
arange()	np.arange(0, 10, 2)	Creates numbers in a specified range.
linspace()	np.linspace(0, 1, 5)	Creates evenly spaced numbers.
ndim	arr.ndim	Shows the number of dimensions.
shape	arr.shape	Shows the size of each dimension.
size	arr.size	Shows the total number of elements.
dtype	arr.dtype	Shows the data type of elements.
reshape()	arr.reshape(2, 3)	Changes the array shape without changing the data.
flatten()	arr.flatten()	Converts a multi-dimensional array into 1D.
ravel()	arr.ravel()	Converts an array into 1D when possible.
transpose()	arr.transpose()	Swaps the array dimensions.
T	arr.T	Gets the transpose of an array.
astype()	arr.astype(float)	Converts array values to another data type.
sum()	np.sum(arr)	Calculates the sum of values.
mean()	np.mean(arr)	Calculates the average.
median()	np.median(arr)	Finds the middle value.
std()	np.std(arr)	Measures how spread out the values are.
var()	np.var(arr)	Calculates variance.
min()	np.min(arr)	Finds the smallest value.
max()	np.max(arr)	Finds the largest value.
argmin()	np.argmin(arr)	Returns the position of the smallest value.
argmax()	np.argmax(arr)	Returns the position of the largest value.
unique()	np.unique(arr)	Returns unique values.
sort()	np.sort(arr)	Returns sorted values.
abs()	np.abs(arr)	Gets the absolute value of each element.
sqrt()	np.sqrt(arr)	Calculates the square root of each element.
exp()	np.exp(arr)	Calculates the exponential of each element.
log()	np.log(arr)	Calculates the natural logarithm.
dot()	np.dot(a, b)	Calculates the dot product; important in ML and linear algebra.
matmul()	np.matmul(A, B)	Performs matrix multiplication.
maximum()	np.maximum(a, b)	Selects the larger value element-wise.
minimum()	np.minimum(a, b)	Selects the smaller value element-wise.
concatenate()	np.concatenate([a, b])	Joins arrays together.
vstack()	np.vstack([a, b])	Stacks arrays vertically.
hstack()	np.hstack([a, b])	Stacks arrays horizontally.
where()	np.where(arr > 0, 1, 0)	Selects values based on a condition.
random.rand()	np.random.rand(3, 2)	Creates random values between 0 and 1.
random.randn()	np.random.randn(3, 2)	Creates random values from a standard normal distribution.
random.randint()	np.random.randint(0, 10, 5)	Creates random integers.
random.seed()	np.random.seed(42)	Makes random results reproducible.
add()	np.add(a, b)	Adds arrays element by element.
subtract()	np.subtract(a, b)	Subtracts arrays element by element.
multiply()	np.multiply(a, b)	Multiplies arrays element by element.
divide()	np.divide(a, b)	Divides arrays element by element.
Broadcasting	a + 10	Applies an operation between compatible array shapes.
Indexing	arr[0]	Gets an element using its position.
Slicing	arr[1:4]	Gets a range of elements.
Boolean filtering	arr[arr > 10]	Filters array values using a condition.

Function / Method	Syntax	Simple explanation
X matrix	X = np.array([[1, 2], [3, 4]])	Stores ML input features as a numerical matrix.
y vector	y = np.array([0, 1])	Stores ML target values as a numerical vector.